CHAPTER 1: INTRODUCTION

1.1 Problem Statement

To develop a Word Counter & Text Analyzer application using Object-Oriented Programming concepts in Java, capable of performing comprehensive text analysis such as counting words, characters, and sentences, and identifying patterns like word frequency and vowel/consonant distribution. The system aims to simplify text analysis tasks that are otherwise tedious to perform manually, and to demonstrate the practical application of Object-Oriented Programming concepts in Java.

1.2 Objectives

The main objectives of the Word Counter & Text Analyzer are:

- To develop a menu-driven application using Java and Object-Oriented Programming concepts.
- To count the number of words, characters, and sentences in a given text.
- To analyze text at the character level, including vowel and consonant counts.
- To identify the longest and shortest words in the text.
- To compute word frequency and display occurrence counts of each word.
- To provide search functionality for locating specific words within the text.
- To implement find-and-replace functionality for editing text.
- To support case conversion (uppercase, lowercase) of the given text.
- To allow analysis of text directly from external files.
- To improve the efficiency and accuracy of manual text analysis tasks.

1.3 Scope of the Project

The scope of this project is limited to basic text analysis and processing operations. The system provides functionalities for text input, statistical analysis, character-level analysis, word frequency analysis, search, find-and-replace, case conversion, and file-based analysis. The project is intended for educational purposes to demonstrate the application of Object-Oriented Programming concepts in Java.

The current version includes:

- Text Input (via console or file)
- Word, Character, and Sentence Counting
- Vowel and Consonant Counting
- Longest and Shortest Word Identification
- Word Frequency Analysis
- Word Search
- Find and Replace
- Case Conversion
- File Analysis

Future versions can include support for multiple languages, graphical visualization of word frequency, readability scoring, and a graphical user interface (GUI).

1.4 Technologies Used

Technology	Purpose
Java	Programming Language
Object-Oriented Programming (OOP)	Software Development Methodology
Java Collections (ArrayList, HashMap)	Storing words and their frequencies
Java String and StringBuilder Classes	Text processing and manipulation
Java File I/O (BufferedReader/FileReader)	Reading text from external files
VS Code / IntelliJ IDEA / Eclipse	Development Environment

CHAPTER 2: REQUIREMENT ANALYSIS

2.1 Functional Requirements

The Word Counter & Text Analyzer should provide the following functionalities:

2.1.1 Text Input

The system should:

- Allow the user to enter or input text for analysis.
- Accept text either through direct console input or via a file (see File Analysis).
- Store the entered text for use across all analysis operations.

2.1.2 Word Count

Counts the total number of words present in the entered text. The system should:

- Takes the text entered by the user.
- Separates the text into individual words using spaces/whitespace.
- Counts the words.
- Displays the total word count.

2.1.3 Character Count

Counts the total number of characters in the entered text. The system should:

- Counts letters, numbers, spaces, and special characters.
- Displays the total character count.
- Depending on the program design, spaces can either be included or excluded.

2.1.4 Sentence Count

The system should:

- The system accepts text from the user and identifies the number of sentences present in the given text.
- It analyzes sentence-ending punctuation such as periods, question marks, and exclamation marks, and displays the total sentence count.

2.1.5 Vowel & Consonent Count

The system should:

- The system analyzes the character by character entered text and identifies vowels and consonants
- It counts and displays the total number of vowels and consonants present in the given text.

2.1.6 Longest & Shortest Word

The system should:

- The system analyzes the entered text and separates it into individual words.
- compares the length of each word to identify the word with the maximum and minimum number of characters.
- The system then displays the longest word and shortest word along with their respective lengths

2.1.7 Word frequency

The system should:

- The system examines all the words present in the entered text and keeps track of the number of times each word occurs.
- It compares the words and calculates their individual frequencies.
- The system then displays the words along with the number of times each word appears in the text.

2.1.8 Search Word

The system should:

- The system allows the user to enter a specific word that needs to be searched.
- It scans the given text and checks for occurrences of the entered word.
- The system then displays whether the word is present and, if found, shows the total number of occurrences of that word.

2.1.9 Find & Replace

The system should:

- Allow the user to specify a target word or phrase to find. Replace all occurrences of the specified word/phrase with a new word/phrase.
- Display the updated text after replacement.

2.1.10 Case Conversion

The system should:

- Allow the user to convert the text to uppercase.
- Allow the user to convert the text to lowercase.
- Allow the user to convert the text to title case (optional enhancement).

2.1.11 File Analysis

The system should:

- Allow the user to load text from an external file for analysis.
- Perform all supported operations on the file's content.
- Optionally save the analysis results back to a file.